

ANATOMY LL 4TH CLASS REVISION

Mr. Solomon

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2018

THE HIP JOINT

CLASSIFICATION OF JOINTS

1. BASED ON STRUCTURE

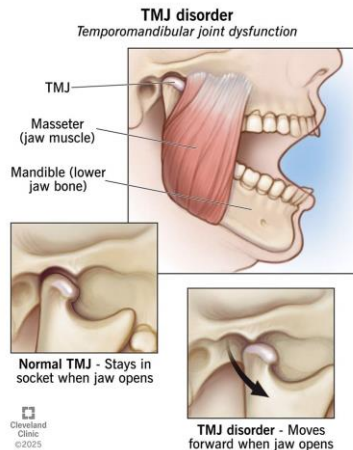
- **Fibrous joints**
 - Bones joined by fibrous tissue
 - No joint cavity
 - No movement
 - Example: Sutures of skull
- **Cartilaginous joints**
 - Bones joined by cartilage
 - Limited movement
 - Example: Intervertebral discs, pubic symphysis
- **Synovial joints**
 - Joint cavity present
 - Articular cartilage + synovial fluid
 - Freely movable
 - **Hip joint belongs to this type**

2. BASED ON DEGREE OF MOBILITY

- **Synarthrosis** - immovable (e.g., skull sutures)
- **Amphiarthrosis** - slightly movable (e.g., pubic symphysis)
- **Diarthrosis** - freely movable
 - **All synovial joints, including the hip joint**

Important note:

- Only one movable joint in the skull →
Temporomandibular joint (TMJ)



MOBILITY VS STABILITY CONCEPT

- Higher mobility = lower stability
- Lower mobility = higher stability

➡ The hip joint sacrifices some mobility to gain stability, because it is a weight-bearing joint.

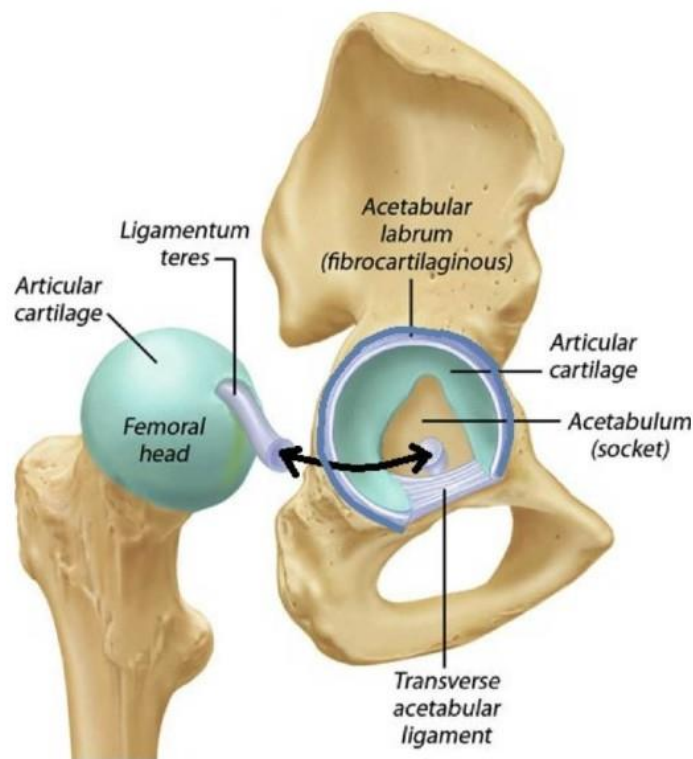
WHY THE HIP JOINT IS STABLE

1. SHAPE OF ARTICULAR SURFACES

- Head of femur is large and spherical
- Acetabulum is deep

◆ Acetabular labrum

- A fibrocartilaginous rim attached to the acetabulum
- Functions:
 - Deepens the socket
 - Increases surface area
 - Holds the femoral head firmly
 - Increases stability without sacrificing too much movement



2. POSITION OF MAXIMUM STABILITY (CLOSE-PACKED POSITION)

- Hip joint is most stable(Naturally) in:
 - Extension
 - Adduction
 - Medial rotation
- Hip joint is least stable in:
 - Flexion
 - Abduction
 - Lateral rotation

✦ This explains why:

- We cannot over-adduct the hip or glenohumeral joint
- Extreme positions increase risk of injury or dislocation

3. EFFECT OF HIP FLEXION

- When the **hip is fully flexed**:
 - Pressure increases on the **anterior part of the joint capsule**
 - Posterior structures become relatively lax
- This makes the joint **vulnerable to posterior dislocation**, especially in trauma

Example:

- Sitting position during a car accident → force transmitted along femur → posterior dislocation of hip

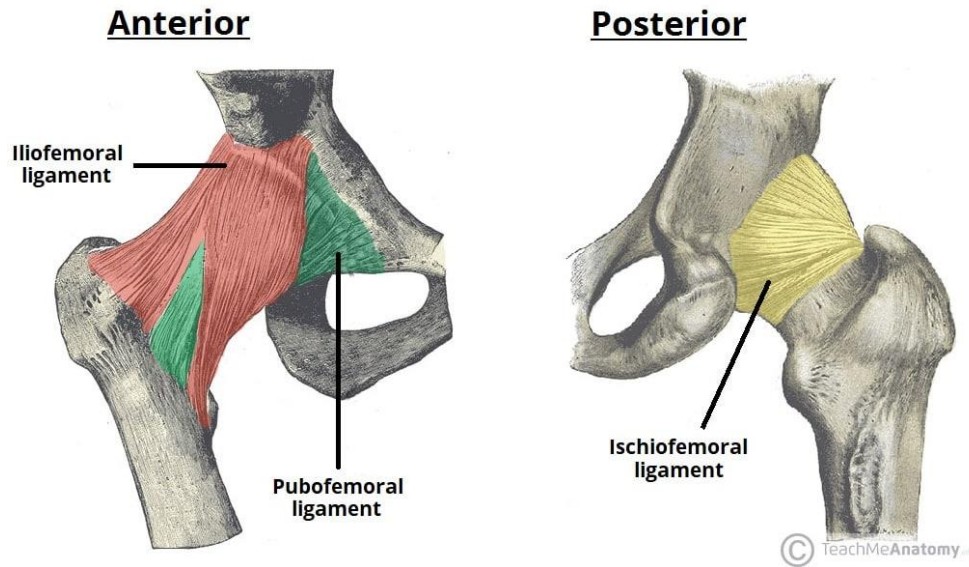
LIGAMENTS OF THE HIP JOINT

FUNDAMENTAL LIGAMENT

- **Iliofemoral ligament** (Y-shaped ligament of Bigelow)
 - Strongest ligament of the hip
 - Prevents **hyperextension**
 - Maintains upright posture with minimal muscle activity

Other ligaments (for context):

- Pubofemoral ligament
- Ischiofemoral ligament

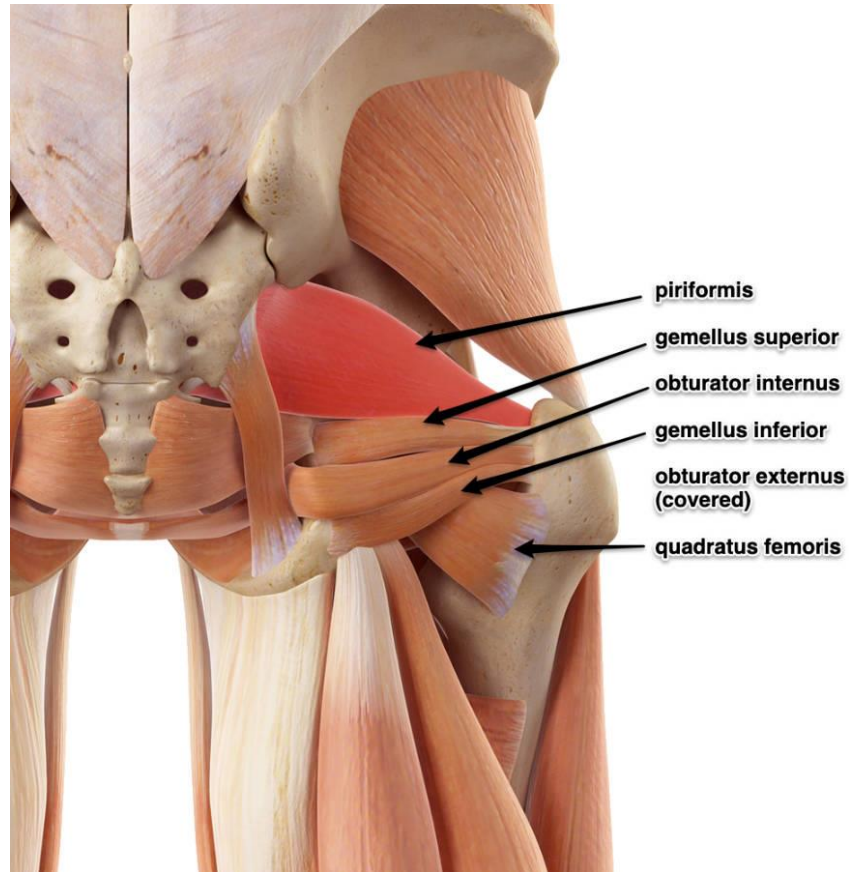


MUSCULAR SUPPORT OF THE HIP (ROTATOR-CUFF-LIKE MUSCLES)

These short lateral rotators stabilize the hip joint, similar to how rotator cuff muscles stabilize the shoulder.

DEEP MUSCLES OF THE HIP

- Piriformis
- Obturator Internus
- Gemellus Superior
- Gemellus Inferior
- Quadratus Femoris



Function:

- Stabilize the femoral head in the acetabulum
- Prevent unwanted movements during walking and standing

DISLOCATION OF THE HIP JOINT

TYPES OF HIP DISLOCATION

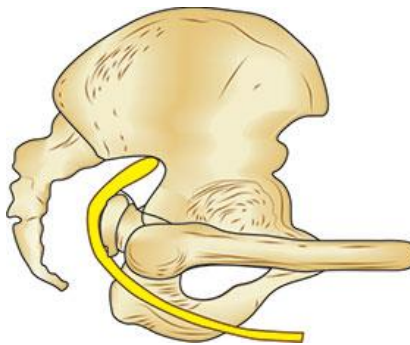
- Anterior
- Posterior □
- Lateral (rare)

POSTERIOR DISLOCATION

- **Most common**
- **Most serious**
- Usually occurs when:
 - Hip is flexed
 - Knee hits dashboard (dashboard injury)

Clinical importance:

- Risk of **sciatic nerve injury**
- Risk of **avascular necrosis of femoral head**



WHY POSTERIOR DISLOCATION OF THE HIP IS VERY SERIOUS

Posterior dislocation is not just common – it is **dangerous** because of **what lies behind the hip joint**.

1. RELATION TO THE SCIATIC NERVE

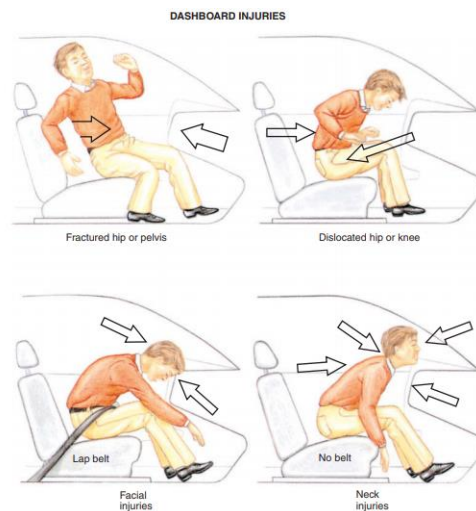
- The **sciatic nerve runs posterior to the hip joint**
- During posterior dislocation:
 - The **head of femur moves backward**
 - It can **stretch, compress, or tear the sciatic nerve**

This is why posterior dislocation is neurologically catastrophic compared to anterior dislocation.

2. COMMON MECHANISM - DASHBOARD INJURY

- Typical scenario:
 - Person sitting in a car
 - Hip flexed, knee flexed
 - Sudden collision → knee hits dashboard
- Force travels **along the femur**
- Femoral head is driven **posteriorly out of the acetabulum**

This exact hip position (flexion + adduction + medial rotation) is the **least stable position of the hip** → perfect setup for posterior dislocation.



3. NO PROTECTIVE CUSHION FOR THE SCIATIC NERVE

- There is **no physiological padding** (no strong muscle or fat layer) between:
 - Gluteal muscles
 - Sciatic nerve
- So when the femoral head dislocates posteriorly:
 - The sciatic nerve takes the hit directly

➡ This explains why **sciatic nerve injury is common and severe**.

EXAMINATION OF SCIATIC NERVE INJURY (POSTERIOR DISLOCATION)

IMPORTANT CONCEPT FIRST

- **Posterior thigh has NO cutaneous (skin) sensory supply**
 - SO, Sensation there is minimal and not reliable for testing
 - Therefore:
 - **Sciatic nerve injury is examined mainly by MOTOR testing**

HOW TO EXAMINE HAMSTRINGS (SCIATIC NERVE)

Hamstrings = semitendinosus, semimembranosus, L.H biceps femoris

→ All innervated by the **sciatic nerve**

STEPS:

1. Patient lies **prone** (on the abdomen)
2. Ask the patient to:
 - Extend the thigh slightly at the hip
 - Flex the knee against resistance
3. Examiner places fingers on the **posterior thigh**
4. Feel for **muscle contraction**

INTERPRETATION:

- **Normal:** Knee flexion present, muscles contract
- **Positive sign (sciatic nerve injury):**
 - No knee flexion
 - No hamstring contraction



EFFECT OF SCIATIC NV INJURY BELOW THE KNEE

- Sciatic nerve divides into:
 - Tibial nerve
 - Common peroneal (fibular) nerve
- Injury to sciatic nerve → **both divisions affected**

➔ Result:

- **Leg and foot become functionally useless**
 - No dorsiflexion
 - No plantarflexion
 - No toe movements

Exception:

- **Anteromedial side of leg**
 - Supplied by **saphenous nerve**
 - Sensory only
 - Comes from femoral nerve
- So sensation here may still be present, but **no motor function below knee**

PATELLAR TENDON REFLEX (L2-L4)

POSITION OF PATIENT

- Patient sits comfortably
- Legs dangling freely
- Muscles relaxed

SITE OF TAPPING

- Tap the **patellar tendon**
- Specifically at the **apex of the patella**

MECHANISM

1. **Stimulus**
 - Sudden stretch of quadriceps muscle
2. **Receptor**
 - **Muscle spindle** (NOT correct but the instructor mentioned **Golgi tendon** organ as example)
 - Muscle spindle detects stretch
3. **Afferent limb**
 - Sensory fibers travel via **femoral nerve**
4. **Spinal cord processing**
 - Segment **L2-L4**
 - Monosynaptic reflex
5. **Efferent limb**
 - Motor fibers via **femoral nerve**
6. **Effector**
 - Quadriceps femoris contracts
7. **Response**
 - **Extension of knee**

Clinical value:

- Tests integrity of:
 - Femoral nerve
 - L2-L4 spinal segments

INGUINAL LIGAMENT

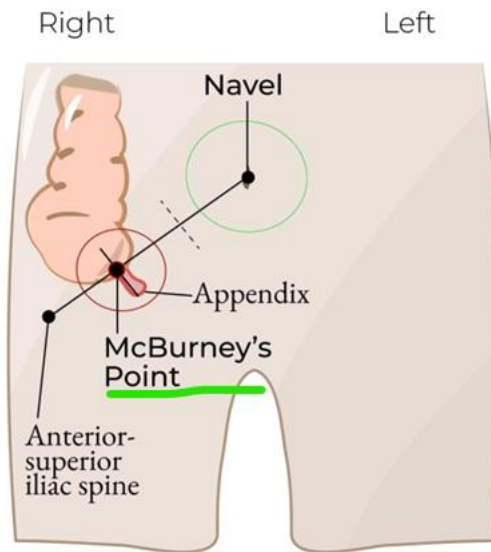
ANATOMY

- Extends from:
 - **ASIS (Anterior Superior Iliac Spine)**
 - **To Pubic tubercle**
- It forms the **base of the femoral triangle**

MCBURNEY'S POINT (APPENDIX LANDMARK)

- Draw a line from:
 - **ASIS → Umbilicus**
- This line is:
 - **Oblique**, not horizontal
- McBurney's point lies at:
 - Junction of **lateral 1/3 and medial 2/3**
- Located in the **right iliac fossa**

McBurney's Point



Tenderness at McBurney's point is a key exam finding in appendicitis.

CLINICAL IMPORTANCE

- **Appendicitis**
 - Pain localizes at McBurney's point
- **Psoas sign**
 - Pain increases during hip flexion
 - Indicates irritation of psoas muscle by inflamed appendix

FEMORAL TRIANGLE

BOUNDARIES

- **Superior (Base):** Inguinal ligament
- **Lateral:** Medial border of Sartorius
- **Medial:** Lateral border of Adductor longus

FLOOR

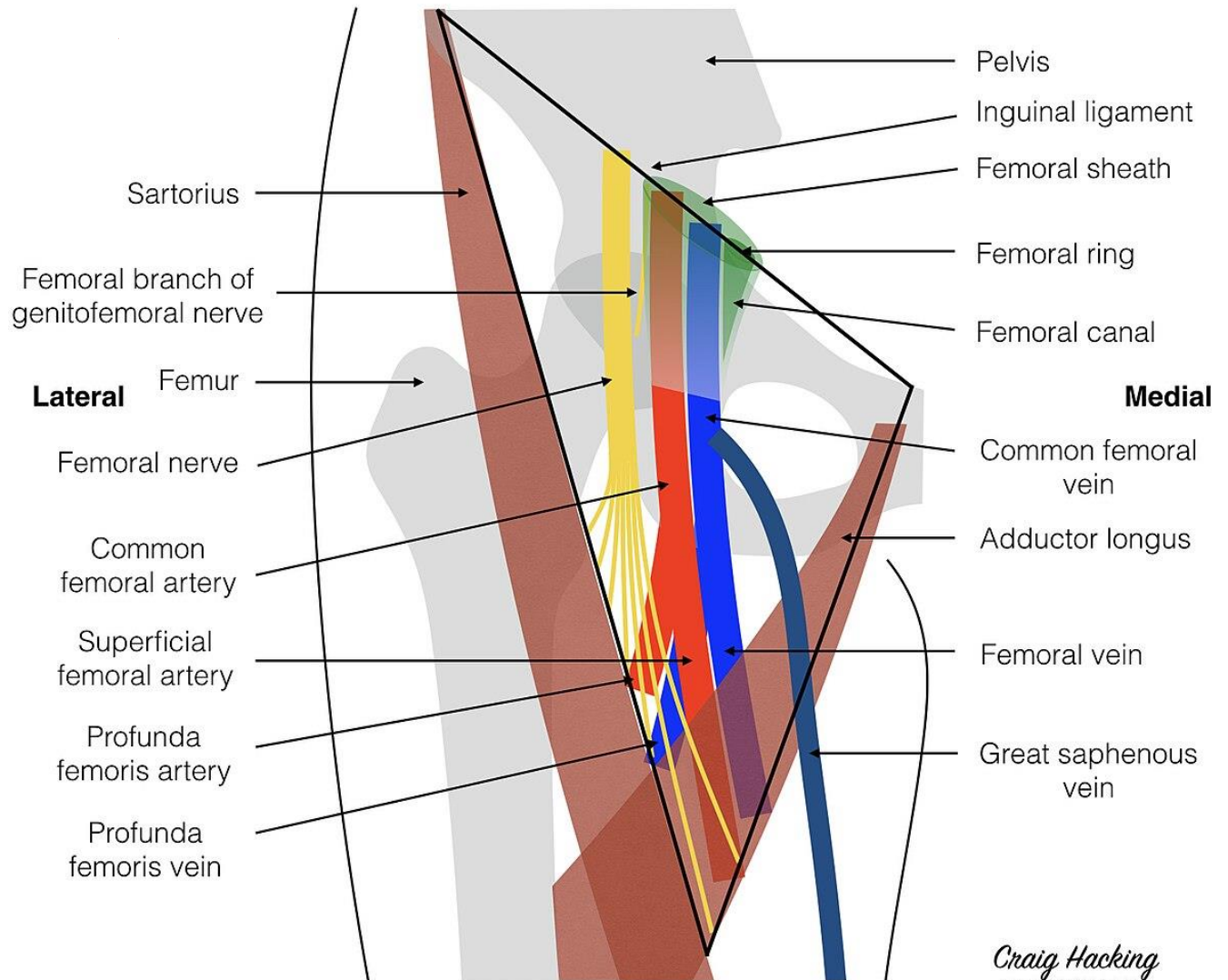
- Iliopsoas
- Pectineus

ROOF

- Skin
- Superficial fascia
- Deep fascia (fascia lata)

“Hematoma due to rupture of profunda femoris artery localizes in the femoral triangle because the artery originates there and the loose fascial planes allow blood accumulation.”

FEMORAL TRIANGLE



Craig Hacking
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WHAT IS THE FEMORAL TRIANGLE?

The **femoral triangle** is a **triangular depression** in the **upper anterior thigh**, just below the inguinal ligament. It is a **gateway region** where major **neurovascular structures** pass from the abdomen into the lower limb.

Think of it as a **traffic junction**: nerves, arteries, veins, and lymphatics all squeeze through here.

BOUNDARIES

- **Base (Superior):** Inguinal ligament
- **Lateral:** Medial border of Sartorius
- **Medial:** Lateral border of Adductor longus(though the correct one is medial border of Adductor canal)

APEX OF THE FEMORAL TRIANGLE

WHAT IS THE APEX?

- The **apex** is the **inferior pointed end** of the femoral triangle
- It is formed where:
 - **Sartorius crosses over Adductor longus**

This crossing narrows the triangle into a funnel.

WHAT HAPPENS AT THE APEX?

- The femoral triangle **continues inferiorly** as the **Adductor (Subartorial) Canal**
- So:
 - **Femoral triangle → Adductor canal**

This continuity is extremely important clinically.

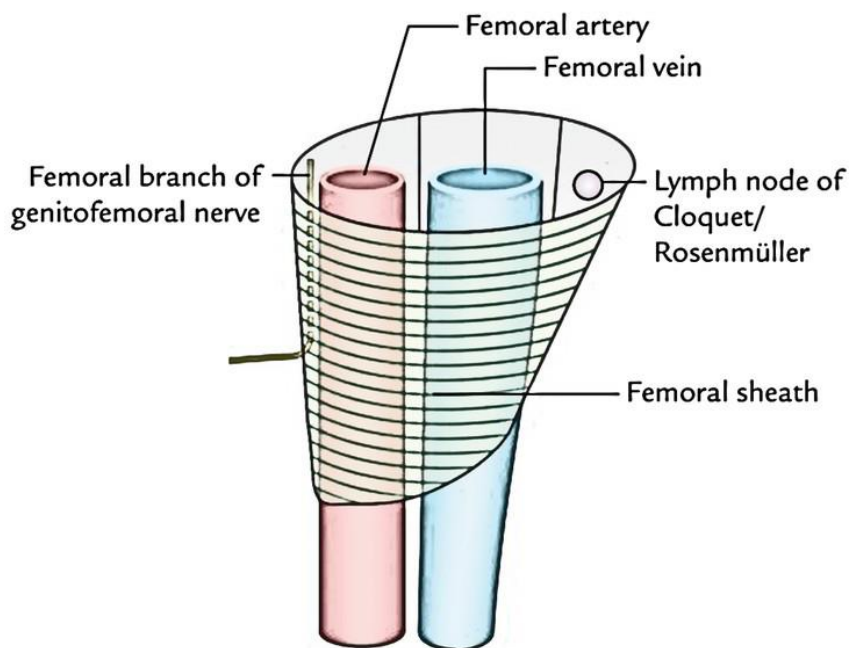
CONTENTS OF THE FEMORAL TRIANGLE (CORE CONCEPT)

The contents are arranged in a **very precise lateral → medial order**, because vessels cannot randomly twist without consequences.

LATERAL → MEDIAL ARRANGEMENT (MNEMONIC: NAVEL)

- **N** - Femoral Nerve
- **A** - Femoral Artery
- **V** - Femoral Vein
- **E** - **Empty space** (Femoral canal)
- **L** - Lymph nodes (deep inguinal)

- ✦ The **femoral nerve** lies outside the **femoral sheath**
- ✦ The artery, vein, and canal are inside the **femoral sheath**



WHAT IS THE FEMORAL CANAL AND WHY IS IT IMPORTANT?

FEMORAL CANAL

- It is the **medial compartment of the femoral sheath**
- Contains:
 - Lymphatics
 - A deep inguinal lymph node **(Cloquet node)**
- Appears “empty” because it does **not** contain artery or nerve

WHY DOES THE BODY NEED A FEMORAL CANAL?

- ✓ Beka Badhasa Speaking: “Because the **femoral vein carries a huge volume of blood** from the lower limb”
- ✓ Mr. Solomon: Excellent!

When venous return increases:

- The femoral vein **expands**
- The femoral canal provides **space for this expansion**

This is not a design flaw – it’s **pressure management**.

“God is the greatest engineer”

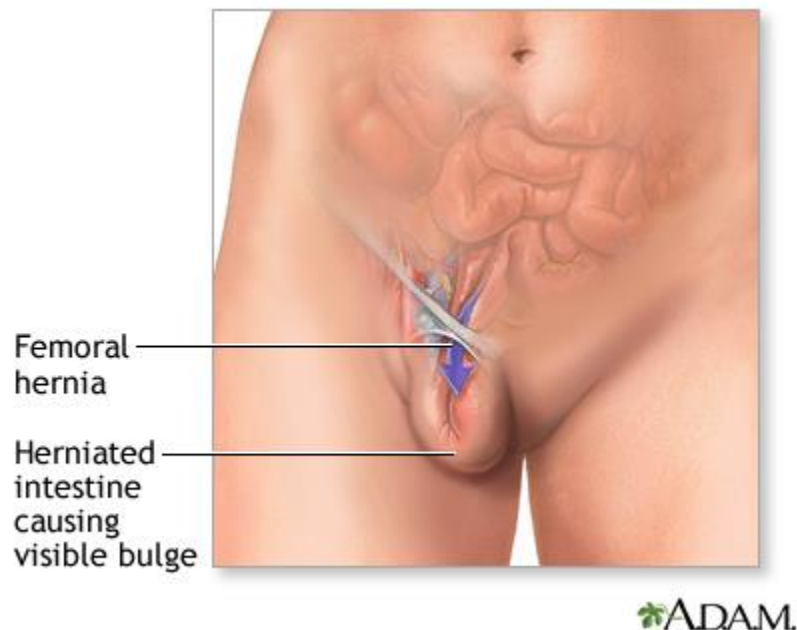
FEMORAL HERNIA (CLINICAL CORRELATION)

WHY FEMORAL HERNIA OCCURS

- The femoral canal is a **potential weak space**
- If the **anterolateral abdominal wall weakens**, abdominal contents may push down

PATH OF A FEMORAL HERNIA

- Abdominal viscera
 - enter **femoral canal**
 - emerge at **saphenous opening**
 - present as a groin swelling



IMPORTANT

- The swelling is felt **below and lateral to the pubic tubercle**

WHY FEMORAL HERNIA IS MORE COMMON IN FEMALES

- Female pelvis is **wider**
- Femoral canal is **broader**
- Increased risk during:
 - Pregnancy
 - Raised intra-abdominal pressure

DIFFERENTIAL DIAGNOSIS OF A GROIN SWELLING

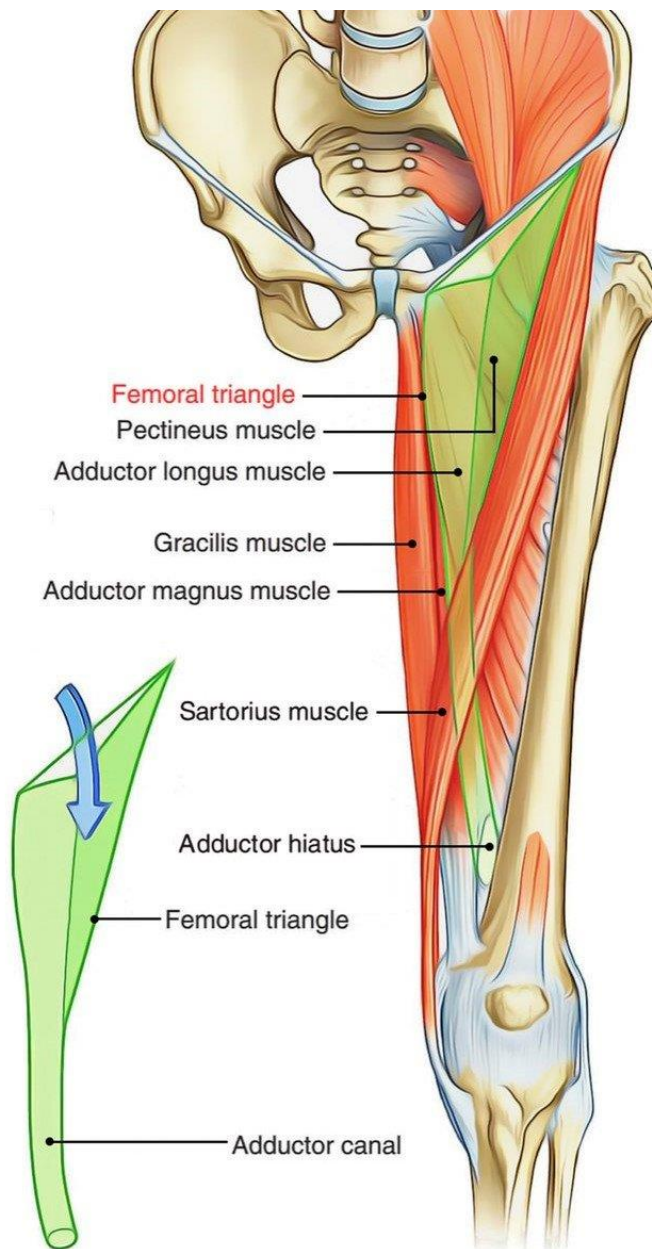
A swelling in this area is **not always a hernia:**

- Femoral hernia
- Great saphenous vein varix
- Enlarged lymph node
- Benign tumor (lipoma)

ADDUCTOR (SUBARTORIAL) CANAL

WHAT IS THE ADDUCTOR CANAL?

- A musculo-fascial tunnel in the middle third of the thigh
- Extends from:
 - Apex of femoral triangle
 - To Adductor hiatus



OPENINGS OF THE ADDUCTOR CANAL

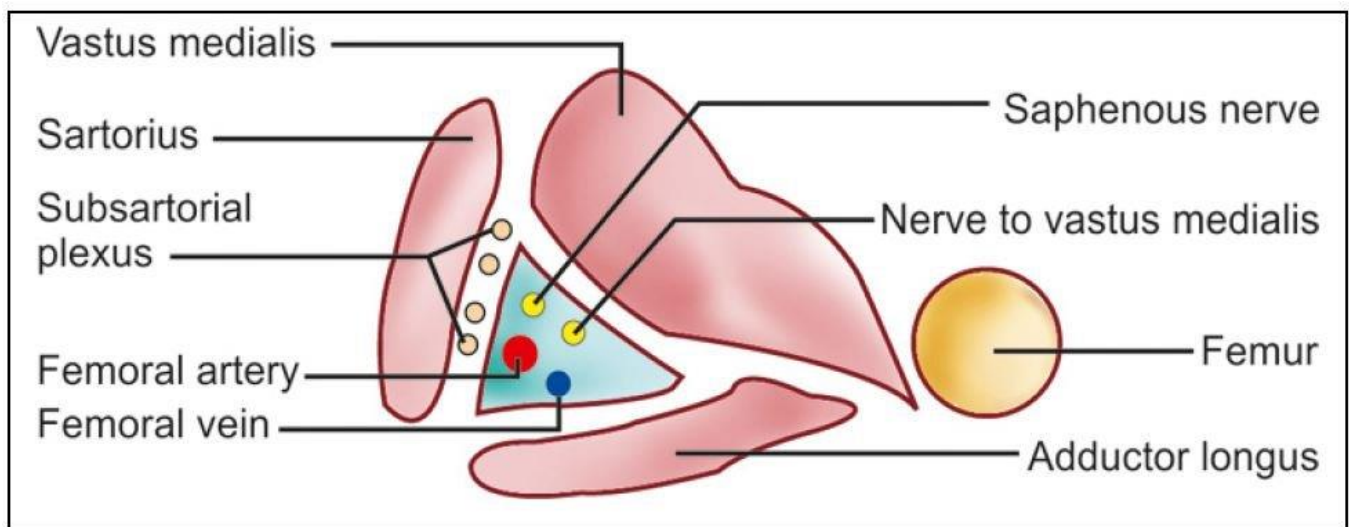
1. **Proximal opening**
 - Apex of femoral triangle
2. **Distal opening**
 - **Adductor hiatus**
 - Leads into **popliteal fossa**

Through this hiatus:

- Femoral artery → becomes **popliteal artery**
- Femoral vein → becomes **popliteal vein**

CONTENTS OF THE ADDUCTOR CANAL

- **Femoral artery**
- **Saphenous nerve**
 - Sensory branch of femoral nerve
- **Nerve to vastus medialis**
- **Femoral vein**



RELATIONSHIP OF ARTERY AND VEIN (ORIENTATION CHANGES)

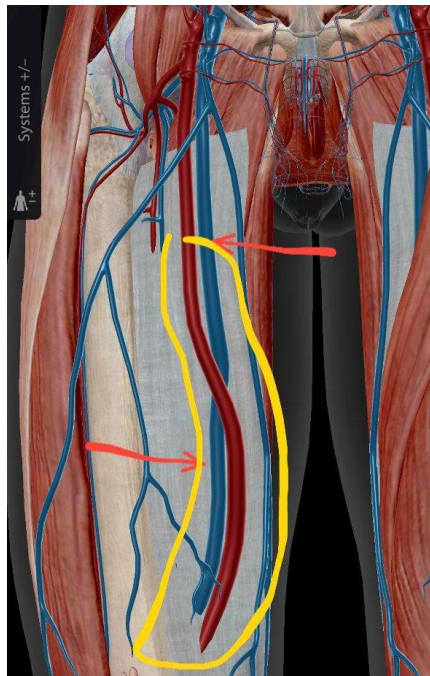
This part confuses us because vessels **rotate around each other** as they descend.

PROXIMALLY (NEAR FEMORAL TRIANGLE)

- Artery → **Lateral**
- Vein → **Medial**

DISTALLY (NEAR ADDUCTOR HIATUS)

- Artery → **more anterior and Medial**
- Vein → **posterior and Lateral**



- In **popliteal artery rupture**, bleeding is massive
- To control it:
 - You clamp the **femoral artery proximally**

ADDUCTOR MUSCLE GROUP

PRIME MOVER OF ADDUCTION

- **Adductor longus** is the main adductor of the thigh



NERVE SUPPLY

- Mostly by **Obturator nerve**
- Origin: **Anterior divisions of L2-L4**
- Enters thigh through **obturator canal**

CLINICAL NOTE

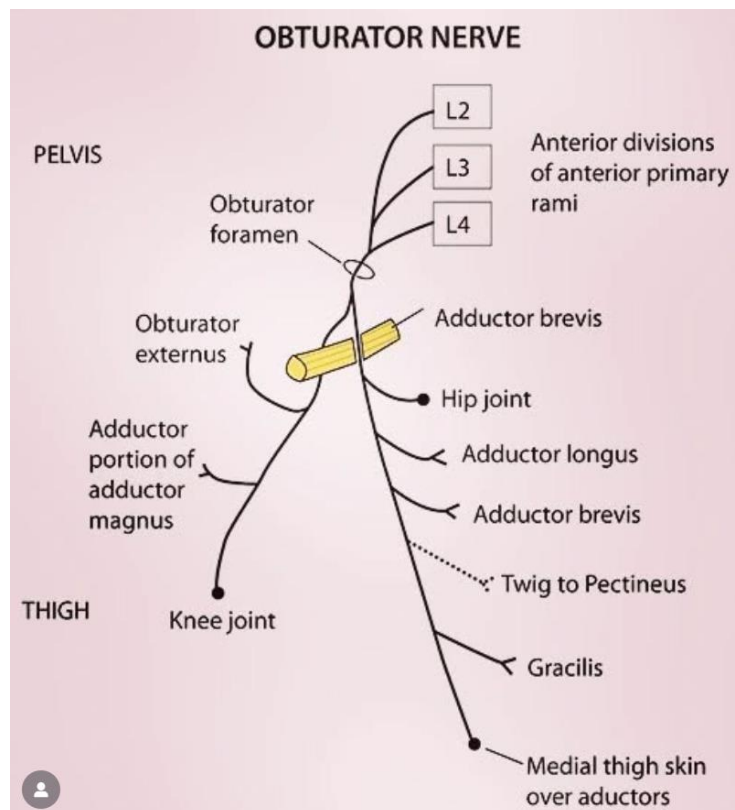
- Injury to adductor muscles causes **groin pain**
- Pain is referred to the groin because of obturator nerve distribution

ADDUCTOR BREVIS - A KEY DIVIDER

- **Adductor brevis** is special:
 - It is pierced by vessels
 - It separates:
 - Anterior compartment
 - Posterior compartment

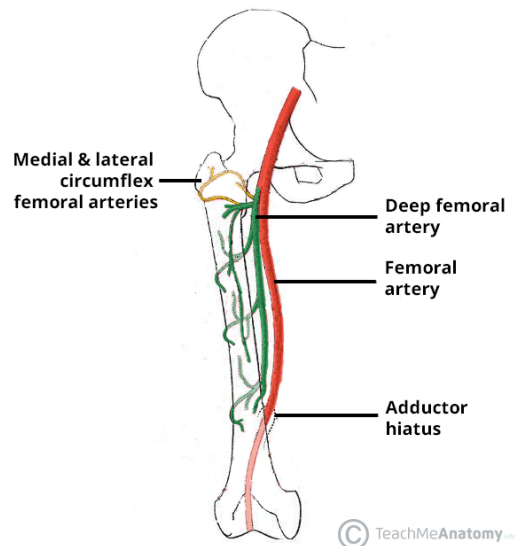
NERVE RELATIONS

- Obturator nerve divides into:
 - Anterior division
 - Posterior division
- These divisions pass around adductor brevis



PERFORATING ARTERIES

- Arise from **Profunda femoris artery**
- Pass through **perforations in adductor magnus**
- Supply the **posterior compartment of thigh**



NOTE:

- ANTERIOR DIVISION OF THIGH: SUPPLIED BY **FEMORAL ARTERY**
- MEDIAL DIVISION OF THIGH IS SUPPLIED BY **OBTURATOR ARTERY**
- POSTERIOR COMPARTMENT OF THIGH SUPPLIED BY **PERFORATING ARTERY OF FEMORAL ARTERY**

THE END OF LECTURE 4